Akoizo Original Practice - Form 3 - Answer key

Division B | Scrambler | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) Doubling wheel radius with the same axle and string does what to driven distance?

A. Doubles it

B. Halves it

C. Quadruples it

D. No change

Correct: A

2. (2 points) What is the Distance Score for a Failed Run?

Unit: Rules and the low-score system.
Worked-case data: A run stops 12.4 cm from the Target Point in 5.31 s with an Inside Can Distance of 60.0 cm and no violations.
Find the Run Score.
Separate investigation observations:
Vehicle A: Stops 20 cm away in 3.0 s.
Vehicle B: Stops 6 cm away in 7.5 s.
Both: No cans, no violations.

A. 2500 points

B. 0 points

C. 150 points

D. 300 points

Correct: A

3. (2 points) Why must the stopping setting change for a curved path?

Unit: The Can Bonus and curved paths.
Worked-case data: Target 9.00 m. The team sets a 60 cm gap. Find the bonus, the gap center offset, and the arc radius.
Separate investigation observations:
Arc runs: Passed between cans 10 of 10.
Stopping: Stopped 18 cm short of the target each time.
Setting: Turns calculated from the straight-line distance.

A. The target moves

B. Curves shrink wheels

C. The arc is longer than the straight-line distance

D. The cans slow the vehicle

Correct: C

4. (2 points) Why recheck calibration on the competition floor?

Unit: Automatic stopping.
Worked-case data: Target 9.25 m. Wheels are 7.50 cm in diameter. The vehicle skids 3 cm after braking, and the egg tip sits 10 cm ahead of the point used as the start reference. How many turns?
Separate investigation observations:
Setting 30.0 turns: Stops at 7.21, 7.24, 7.19 m.
Setting 35.0 turns: Stops at 8.40, 8.43, 8.38 m.
Target: 8.00 m.

A. Wheels grow

B. The target changes the rules

C. Surface changes skid distance and wheel grip

D. Gravity changes

Correct: C

5. (4 points) List three places energy is lost between the mass and the wheels.

answer: Friction in pulleys and bearings, motion of the falling mass itself, and string stretch or slip. (4 points).

6. (4 points) Why build calibration tables for both straight and bonus runs?

Unit: Calibration and competition day.
Worked-case data: Table: 30 turns -> 7.20 m, 34 turns -> 8.15 m, 38 turns -> 9.10 m. Set for a State target of 8.60 m.
Separate investigation observations:
Run 1: Straight run: 4.2 cm from target, 6.1 s, no cans.
Run 1 score: 100 + 8.4 + 6.1 = 114.5.
Option for Run 2: 60 cm can gap; bonus runs average 9 cm error, 6.5 s.

answer: Curved paths are longer and behave differently, so each needs its own settings. (4 points).

7. (4 points) Why is a progressive slowdown better than a sudden stop?

Unit: Protecting the egg.
Worked-case data: A vehicle arrives at 1.5 m/s and stops in 6 cm. Find the average deceleration in g.
Separate investigation observations:
Design A: Arrives 3.0 m/s, stops in 2 cm; 2 of 5 eggs cracked.
Design B: Arrives 1.5 m/s, stops in 5 cm; 0 of 5 cracked.
Question: Compare decelerations.

answer: Spreading the stop over a longer distance reduces deceleration and egg stress. (4 points).

8. (4 points) Why must wheel diameter be measured precisely?

Unit: Drivetrains: turning a drop into distance.
Worked-case data: A vehicle has 0.50 cm axle radius and 4.0 cm wheel radius. The mass pulls 0.75 m of string. About how far can the vehicle be driven?
Separate investigation observations:
Run 1: String wound in one neat layer: 8.42 m.
Run 2: String piled in three layers: 7.61 m.
Same: Mass and drop height.

answer: Calibration converts axle turns to distance using circumference; a small diameter error accumulates over meters. (4 points).

9. (4 points) How is the final score determined from two runs?

Unit: Rules and the low-score system.
Worked-case data: A run stops 12.4 cm from the Target Point in 5.31 s with an Inside Can Distance of 60.0 cm and no violations. Find the Run Score.
Separate investigation observations:
Vehicle A: Stops 20 cm away in 3.0 s.
Vehicle B: Stops 6 cm away in 7.5 s.
Both: No cans, no violations.

answer: The better of the two Run Scores plus any Final Score penalties. (4 points).

10. (4 points) Why master straight runs before the bonus?

Unit: The Can Bonus and curved paths.
Worked-case data: Target 9.00 m. The team sets a 60 cm gap. Find the bonus, the gap center offset, and the arc radius.
Separate investigation observations:
Arc runs: Passed between cans 10 of 10.
Stopping: Stopped 18 cm short of the target each time.
Setting: Turns calculated from the straight-line distance.

answer: Straight runs build a reliable drivetrain and calibration; adding the curve later isolates new errors. (4 points).

11. (6 points) Solve this case and explain your reasoning: Target 9.25 m. Wheels are 7.50 cm in diameter. The vehicle skids 3 cm after braking, and the egg tip sits 10 cm ahead of the point used as the start reference. How many turns?

reasoning: Circumference = π x 7.50 about 23.56 cm.
Distance to cover before braking = 925 - 3 - 10 = 912 cm.
Turns = 912 ÷ 23.56 about 38.7 turns. (3 points).

conclusion: Set the nut for about 38.7 turns, then confirm on the competition floor. (3 points).

12. (2 points) Which setup is better?

Supplied synthetic observations:
2.0 kg mass: Top speed 4.2 m/s; egg cracked on stop in 3 of 5 runs.
1.2 kg mass: Top speed 2.6 m/s; egg safe 5 of 5; time +1.8 s.
Target: 8.00 m in both cases.

A. 2.0 kg mass

B. No difference

C. 1.2 kg mass

Correct: C

13. (2 points) Should the team try the Can Bonus on Run 2?

Supplied synthetic observations:
Run 1: Straight run: 4.2 cm from target, 6.1 s, no cans.
Run 1 score: 100 + 8.4 + 6.1 = 114.5.
Option for Run 2: 60 cm can gap; bonus runs average 9 cm error, 6.5 s.

A. No; Run 2 replaces Run 1

B. Yes; the expected score is lower and Run 1 is banked

C. No; the bonus never helps

Correct: B

14. (2 points) Define “Backstop” in the context of Scrambler.

answer: Wooden block at least 1.1 cm thick holding the dowels. (2 points).

15. (2 points) Define “Distance ratio” in the context of Scrambler.

answer: Wheel radius divided by axle radius. (2 points).

16. (2 points) Define “Vehicle Distance” in the context of Scrambler.

answer: Point-to-point distance from egg tip to Target Point. (2 points).

17. (2 points) Define “Inside Can” in the context of Scrambler.

answer: Placed by competitors, 0-100 cm from the Outside Can. (2 points).

18. (2 points) Define “Brake” in the context of Scrambler.

answer: Device that stops wheel or axle rotation. (2 points).